Linking Marketing Campaign With University Course Enquiries: A Case Study of University of Hull

Abstract

University student recruitment is an increasingly important issue in UK universities. This study uses the datasets consisting of 146 rows and 33 columns from the marketing department of the University of Hull to discover the most effective marketing campaign generating leads for the institution.Universities as business entities need to reduce cost and increase profit hence the significance of this study which helps to identify the campaign strategy with least cost and most effect. Data cleaning, wrangling , analysis and visualisation were carried out using python. Several researches have focussed on the best campaign strategy to market higher education but many have left out the effect of COVID 19 on University marketing campaigns which the study seeks to cover. Findings from the study reveals that marketing campaign strategy for University of Hull changed during COVID 19 and there is the re-emergence of the need for intensifying telephone marketing which ironically, is the most effective campaign strategy despite having the least input.

Introduction/Background

Marketing campaigns are the plan of actions laid out by businesses to accomplish particular marketing goals such as increase revenue and/or web traffic . They are a set of strategies geared towards achieving an ultimate goal therefore they can be referred to as a means to an end, so it is relevant and important for any organisation to have marketing goals. There is a new reality of organisations needing to identify and assess what marketing campaigns to embark on and the academia is not excluded in this situation. Universities as a business organisation need to overcome the long existing challenge of using wrong or ineffective marketing strategies to create student course enquiries which impacts revenue.

The basic goal of any business entity is to make as many people as possible interested in their products or services and this is regarded as lead generation. Linking marketing campaigns to University course enquiries is basically referring to generating leads/student's interest for University programs. Lead generation has been defined by Świeczak and Lukowski (2016) as the marketing process of involvement and capture of interest in a product or service which is aimed at developing sales plans and, as a consequence, soliciting new clients. The core service Universities provide is knowledge provision and the primary consumers are students. In this study, our case study is the University of Hull, United Kingdom.

Current approaches to data analytics of university marketing have partially solved the problem of market need identification but more is still left undone in terms of prediction/forecasting most effective campaign to embark on. Bridging this gap is what this study seeks to contribute to, thereby making it significant. An overview of this study shows introduction , case study, aim and

objectives in the first section one, section two focuses on literature review and theoretical framework. In section three, we describe data used, method of analysis (methodology), data cleaning and wrangling . Main analysis of research as well as discussion of results is done in section four while section five is for conclusion and recommendation for decision making.

Aim and Objectives

The study aims to examine, model and predict best performing marketing campaign used by the University of Hull in order to optimise resource utilisation and it has the following objectives:

1. To identify the existing marketing campaigns/platform for University of Hull yielding students' interest.

2. To determine and evaluate the performance level of each campaign strategy.

4. To draw a model for the existing campaign strategy.

4.To predict and suggest the best campaign strategy to yield students' interest

Case Study

The case study for this research is the University of Hull (UoH) with its main campus located along Cottingham road in Kingston upon Hull, a city in the East Riding of Yorkshire, England. According to the annual report of 'Our Impact' in 2015, the total number of students at the University of Hull is 16, 393. With a breakdown of full-time undergraduate(UG) student population of 12,068 (74%), Part-time UG - 1,988 - (12%),Full-time postgraduate (PG) - 1,471 (9%) and Part-time PG - 866 (5%). It is therefore important for University of Hull to increase the efficiency of its student recruitment process in order to stay ahead of competition.

Literature Review

Higher Education (HE) marketing surfaced around the mid 1980s as a branch of HealthCare marketing (Hayes, 2007), it became important to accept the idea considering the fact that universities started competing for fundings and more students were needed (Drummond, 2004). That was why Pasternak (2005) asserted that, 'In modern days, Universities are being marketed just like any other products or services'. Therefore we can make use of the same marketing mix-model for marketing regular products to build our marketing strategies for University

campaigns. Filip (2012) added that universities ought to employ divergent communication strategies to hit a specified target audience and these strategies are advertising, sales promotion, public relations and personal selling. Ozbal & Konakli (2015) supported this by saying 'since the key publics are the younger generations (and their relatives), higher education institutions should perform marketing with the latest trends and should be global players'. Ozbal & Konakli (2015) views are in line with Fuchs' (2010) position that the surfacing of social media brought about reorganised communication and has emerged the need for more guidance to the communicators. Digital technology can increase leads e.g leads for Algonquin College's increased by 71% annually by utilising 'Marketo' to streamline their print and digital marketing strategies.

Over the past few years, social media system have therefore grabbed the attention of universities, and past studies have shown that a significant amount of public and private universities and colleges have employed the intense usage of social media platforms (Merrill, 2011; Jan, & Ammari, 2016; Hossain, & Sakib, 2016; Nazeer, 2017). Paladan (2018) investigated the frequent social media networks used by the top 25 universities and colleges in Africa and Asia, and discovered that they make use of popular social media platforms such as Twitter, Facebook, Youtube, and LinkedIn, to promote their undergraduate and postgraduate programs and to disseminate general information .

Clark, Fine, andScheuer (2017) studied the impact of social media on relationship quality between the university and students and results show students following universities' social media handles have better understanding and relationship with universities. Also,the works of Rutter, Roper, and Lettice (2016) and of Zickar, Ron, and Arnold (2018)supports the claim of positive impact of social media on university course enquiry and recruitment. Using artificial intelligence to automate the marketing campaign process is rare in research work and this is the gap this study seeks to bridge. Also, The current literature on COVID-19 and Higher Education Institutions (HEIs) focuses on student mobility and experiences (Chirikov and Soria, 2020; Firang, 2020), teaching and learning (Ahlburg, 2020; Birrelll, 2020; Schleicher, 2020), and the financial viability of HEIs (Carnegie et al., 2022; Thatcher et al., 2020). There is still much unknown about the most effective campaign strategy used for recruiting university students during this period.

Theoretical Framework

The main theory we shall examine is the Marketing-mix theory which gained popularity in academia in 1964, after Borden's publication on 'the concept of marketing-mix'.There were so many elements in the initial mix until McCarthey streamlined it into just 4Ps; product, price, place and promotion. So marketing strategies were based on these 4Ps until questions began to arise in literature regarding its applicability to the service industry in which the education market belongs (Palmer, 2001).Kotler and Fox(1995) further contributed by removing 'product' and replacing it with 'process', 'program' , 'physical facilities' and 'people' thereby making it 7Ps.

Another important theory is Personal selling which according to the American Marketing Association (AMA) is simply the oral presentation in a conversation with one or more prospective purchasers for the purpose of making a sale. Also explaining the personal selling theory further is the AIDAS theory of selling which was first introduced by E.St. Elmo Lewis in 1898 and it believes that the potential buyer's mind passes through the stages of Awareness, Interest, Desire, Action and Satisfaction (AIDAS).

Data

The kind of data used for this study shall be a private and secondary form of data source.This data is provided by the University of Hull and each row represents a different launch of an advertising campaign. It is a record of marketing efforts or campaign strategies used to generate impressions as well as actions about the University's programs. Some of the main features of this data include the project identification number, the platform used to convey the message, targeted audience, campaign goal, advert period, budget and effect. The data consists of 146 rows and 33 columns which we shall be analyzing for both Autumn and Spring period of year 2021 and 2022 respectively. This period lapse with the COVID 19 period and this has forced higher education institutions (HEIs) worldwide to alter their operations with significant impacts on students and institutions (Altbach and de Wit, 2020). Hisel and Pinion (2020) asserted that with the increasing concern over university budgets, especially after the abrupt disruption COVID 19 response caused in spring of 2020, it became more important now than ever to reach out to prospective students.

The first line of action is to get the data cleaned to prepare it for proper usage. The 33 columns (see *Appendix 1*) represent titles and description of variables from which we have 13 floats, 4 integers, 15 objects and 1 bool. Some columns needed to be dropped because their content is not useful and significant to our analysis. Such columns include: 'campaign-id' and 'project_id' whose characters are all unique; 'tags''- which contains empty lists ("[]"); 'alwaysOnActivity' and 'pricingType' has only one value which is false; 'note' which has only 2 not NaN. And 'kpi_conversion_id' which has 12 not NaN. tactic_name is dropped because almost all its features are unique. kpi-percentage and kpi_performance are the same except that kpi_percemtage represents 0.0 with NaN therefore kpi_percentage is dropped.

Outliers are data objects that diverged conspicuously from the direction/trend of other data objects in the dataset. Essence of cleaning is to drop the outliers in order to avoid distortions in our results and the effect of this on our result is insignificant. After cleaning, the relevant column left includes the campaign identification number, Medium platforms , Title/Name of the advertisement program, The completion status of the advert, Expected Response from Audience,

The total advertising budget, Currency in which budget is made, The leftover/unspent budget, Advertisement start and end dates, Rate of effectiveness of advertisement, Advertisement objective, Number of reach and Number of Click generated. The marketing platforms used include: UCAS (Universities and Colleges Admissions Service), The Guardian, Youtube, Snapchat, Rezonence, The Student room, TAN Media, Instagram, Facebook family and Google. Our analysis shall be classification based where we shall identify and assign categories to these data for more accurate analyses and prediction.

Methodology

The dataset used was a combination of 2 smaller datasets in 2021, which are;

- 2021- University of Hull Autumn Open Days which contains 99 rows and 33 columns including title, and
- Hull Clearing and Open Day 2021 dataset, containing 49 rows and 33 columns(having the same title)

Prasad Patil (2018) described Exploratory Data Analysis(EDA) as 'the critical process of performing initial investigations on data so as to discover patterns,to spot anomalies,to test hypotheses and to check assumptions with the help of summary statistics and graphical representations'. EDA reveals some columns consist of empty values(e.g ‘tags’), some have the same value and were reduced to 1(e.g kpi-percentage was dropped for kpi_performance since both contain same value) and some consist of unique figures in integer form which is actually meant for identification and not a representation of the value of the figures they reflect(e.g campaign_id, project_id) so they were removed so as not to alter result accuracy. Column such as ‘AlwaysOnActivity’ was dropped because it contains one value and analysing a single value will only yield a known result so it is insignificant.

There is no way we can effectively use EDA without complementing it with Confirmatory Data Analysis (CDA) to prove accuracy of decisions taken and we propose using non-parametric supervised learning methods to classify marketing campaigns as the independent variables and students' interest as the dependent variable. This method is referred to as decision tree technique it is put to performance to analyse the existing different marketing campaigns and come predict the one with the highest vote in terms of students' interest generator.Exploratory data analysis is first performed on the data to understand each unique feature in the dataset, which is followed by Data Wrangling and feature engineering. From there we build our train and test data sets. `

Data Wrangling

The data wrangling process of the dataset that consisted of 146 rows was cleaned by: to ensure that no duplicate record existed in the dataset, and no duplicate record was present;
The dataset had a total of 52 missing values that were spread across 6 columns (kpi_to_date: 3, impressions: 7, clicks: 4, ctr: 14, cpc: 11, cpm: 13) and they were filled up using the median value of each row. This approach was employed in order to avoid adding any form of skewness that was present in the dataset. Highly correlated features (as seen in figure 3) and other redundant features in the form of a duplicate feature that was present in the dataset were dropped. This figure shows the money spent feature and budget feature are highly correlated. Insights such as this one were used to make informed decisions on which features to drop from the dataset in order not to introduce any form of bias in the model. The dataset was further wrangled to remove outliers by accessing the box plots of each feature.

Feature Engineering

New features were created in the dataset through several techniques of feature engineering. The dates were split into months and days and categorical features were encoded with integer representations and they were scaled between a range of 0 and 1.

Modeling

The dataset was split into 80% train set and 20% validation set. Two approaches were used for the modeling of the dataset. The task was addressed both as a regression and a classification problem. In order to convert the target variable (kpi performance) into a discrete feature, a custom range was defined to create a total of 4 classes. The classes defined had the following ranges:

Range 1: 0.238% - 59.48%
Range 2: 59.48% - 118.96%
Range 3: 118.96% - 178.44%
Range 4: 178.44% - 237.92%

The regression algorithms used include:

- **Linear Regression:** we use this to predict a variable value(dependent variable) from another variable value(independent variable).
- **Ridge Regression:** Since we have a lot of correlated independent variables in the data, there is multicollinearity and this can be corrected by Ridge Regression. This method estimates the coefficients of multiple-regression models on the highly correlated independent variables.
- **Lasso Regression:** This is a method of variable selection and regularization to increase predictability, accuracy and interpretation of result.

- **Decision Tree:** This is a style of deciding by using a tree-like model of decisions and their possible outcomes. It is a data analysis technique that splits our data into several possible entities concerning a specific parameter.
- **Random Forest:** From its name, Random Forest comprises a large number of decision trees agglomerated as a bunch(forest) . Each single tree in the random forest depicts a class prediction and the class with the most votes becomes our model's prediction.
- **XGBoost:** XGBoost means "Extreme Gradient Boosting", and "Gradient Boosting" originates from the paper *Greedy Function Approximation: A Gradient Boosting Machine*, by Friedman(2001). XGBoost classifier is a Machine learning algorithm that is applied for structured and tabular data. It is an implementation of gradient boosted decision trees designed for speed and performance. It is an open-source software library which provides a gradient boosting framework for C++, Java, Python, R, Julia, Perl, and Scala. *Chen & Chen (2016).*
- **LGBM Regression:** This is a gradient boosting decision tree that is highly efficient and reduces memory usage. It is useful for ranking, classification and other machine learning tasks.

  The above algorithms were algorithmic methods used in course of analysing the dataset for this research and were evaluated using the Mean Absolute Error. The classification algorithms used also include Decision Tree, Random Forest, LGBM and XGBoost stated above, as well as:
- **Naive Bayes:** This is a probabilistic classifier based on probability models that incorporate strong independence assumptions.
- **Support vector machine:** are a set of supervised learning methods used for classification, regression and outliers detection.
- **Catboost:** CatBoost is an algorithm for gradient boosting on decision trees that was developed by Yandex researchers and engineers. Its usage spans through search, recommendation systems, personal assistant, self-driving cars, weather prediction and many other tasks.. It is open-source and available for usage by anyone.

and all these algorithms were evaluated using precision, recall and f1 score.

Results

This section shows the outcome from running the algorithms as well as the discussion of results. Figure 1 shows the result of the correlation of the entire dataset before any feature engineering step was performed on the dataset. It shows the money spent feature and budget feature are highly correlated. Such Insights were used to make informed decisions on which features to drop from the dataset in order not to introduce any form of bias in the model.

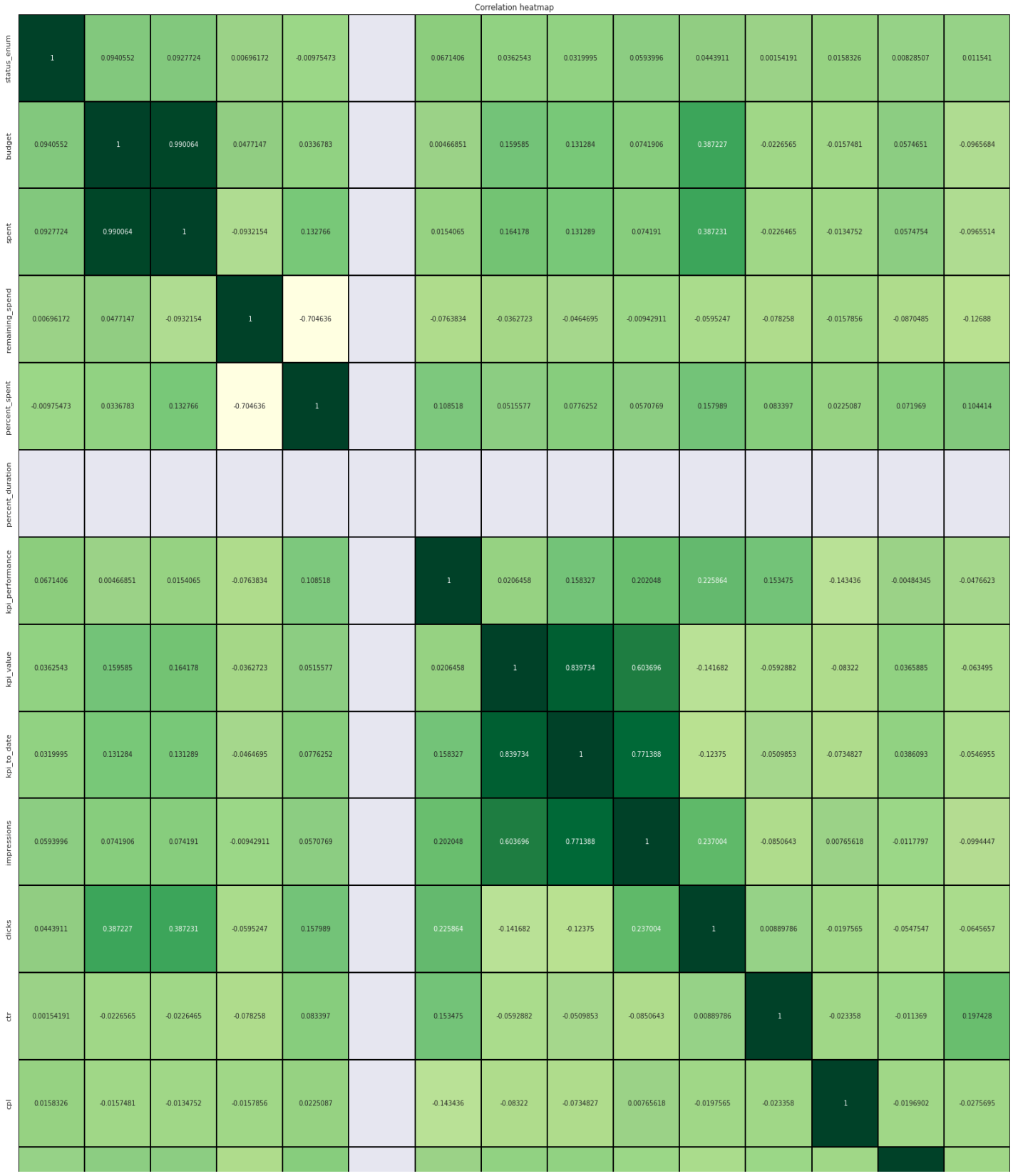

Figure 1: Correlation heatmap plot of the dataset before the feature engineering step.

The table below (Table 1) is a cumulative report of the mean absolute error of all the models created using the regression algorithms earlier mentioned in the Methodology section. From the table, it can be seen that the Decision Tree Model had the best performance with the lowest mean absolute error.
Following this, Table 2 also displays the cumulative result of all the models built with classification algorithms.

Table 1: Results of approaching the task as a Regression problem

S/N	Model	Mean Absolute Error
1	LinearRegression	56.6
2	Ridge	54.9
3	Lasso	56.6
4	DecisionTreeRegressor	**36.6**
5	RandomForestRegressor	41.6
6	XGBRegressor	37.9
7	LGBMRegressor	42.2

Table 2: Results of approaching the task as a classification problem

S/N	Model	F1 Score	Precision	Recall
1	DecisionTreeClassifier	0.59	0.65	0.65
2	RandomForestClassifier	0.55	0.61	0.61
3	GaussianNB	0.34	0.40	0.40
4	SVC	0.43	0.63	0.63
5	CatBoostClassifier	0.60	**0.68**	**0.68**
6	XGBClassifier	**0.61**	0.66	0.66
7	LGBMClassifier	0.52	0.53	0.53

From table 2, XGBoost classification algorithm achieved the highest f1 score with 0.61 whereas Cat boost algorithm achieved the highest recall and precision of 0.68 and 0.68 respectively.

Other techniques like cross-validation training and hyperparameter optimisation were applied with the intent of increasing the model's performance but no positive result was obtained.

Discussion

Extrapolating from the entire working it is interesting to note that the primary KPI used for the evaluation of the marketing campaigns indicated a somewhat inverse result. Of all the primary KPIs employed, the least used was placing of calls but it was the most effective strategy as seen in figure 3a and 3b.

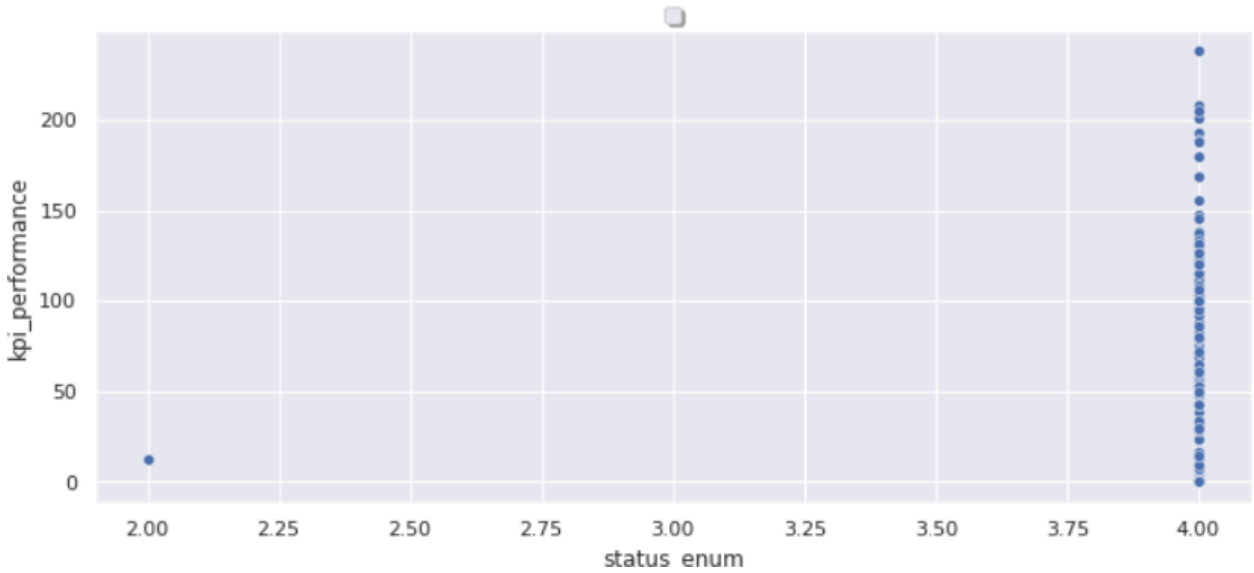

Figure 2a

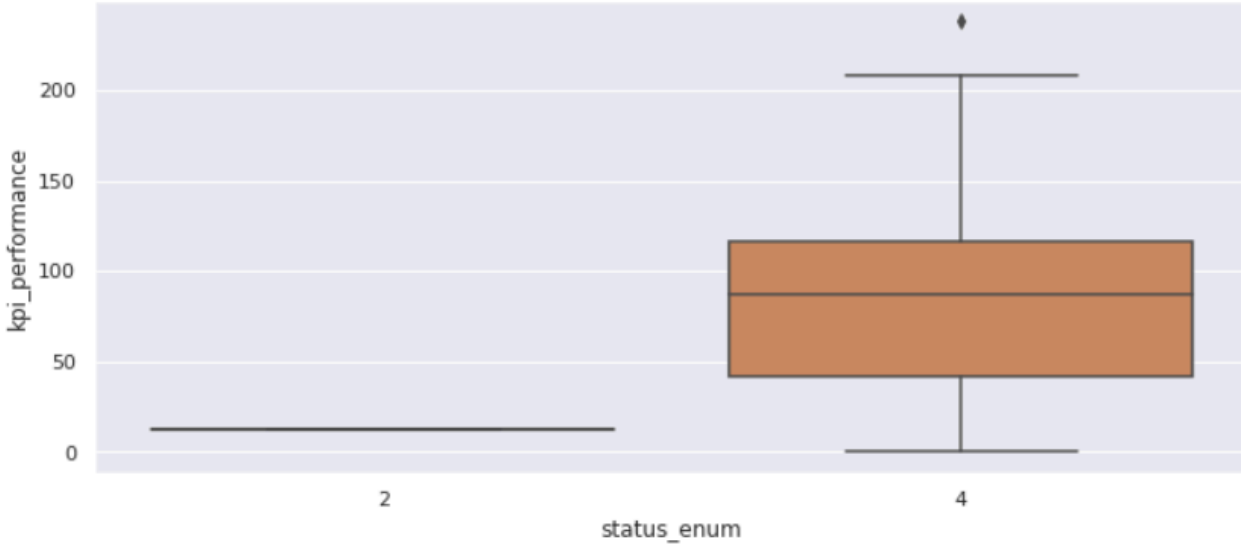

Figure 2b

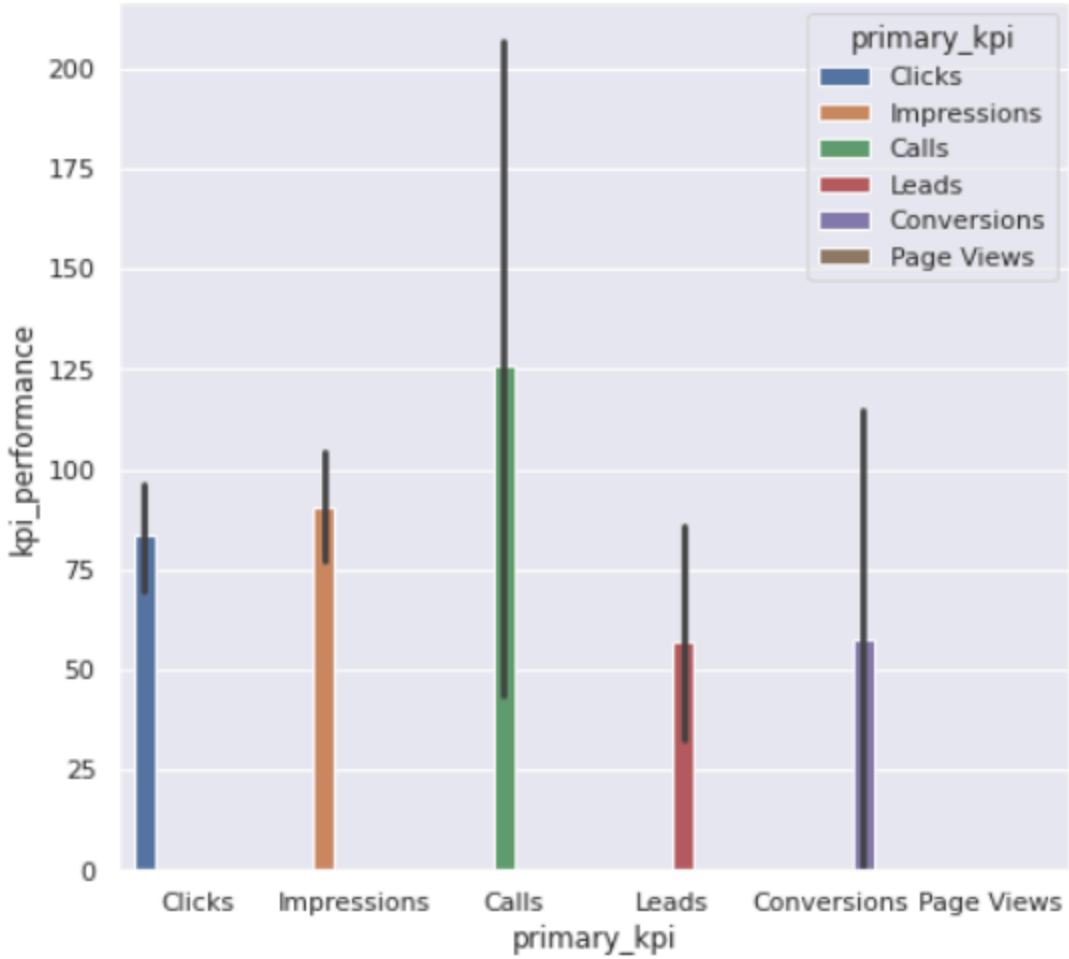
primary_kpi
Clicks
Impressions
Calls
Leads
Conversions
Page Views
kpi_performance
0
25
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Figure 3a

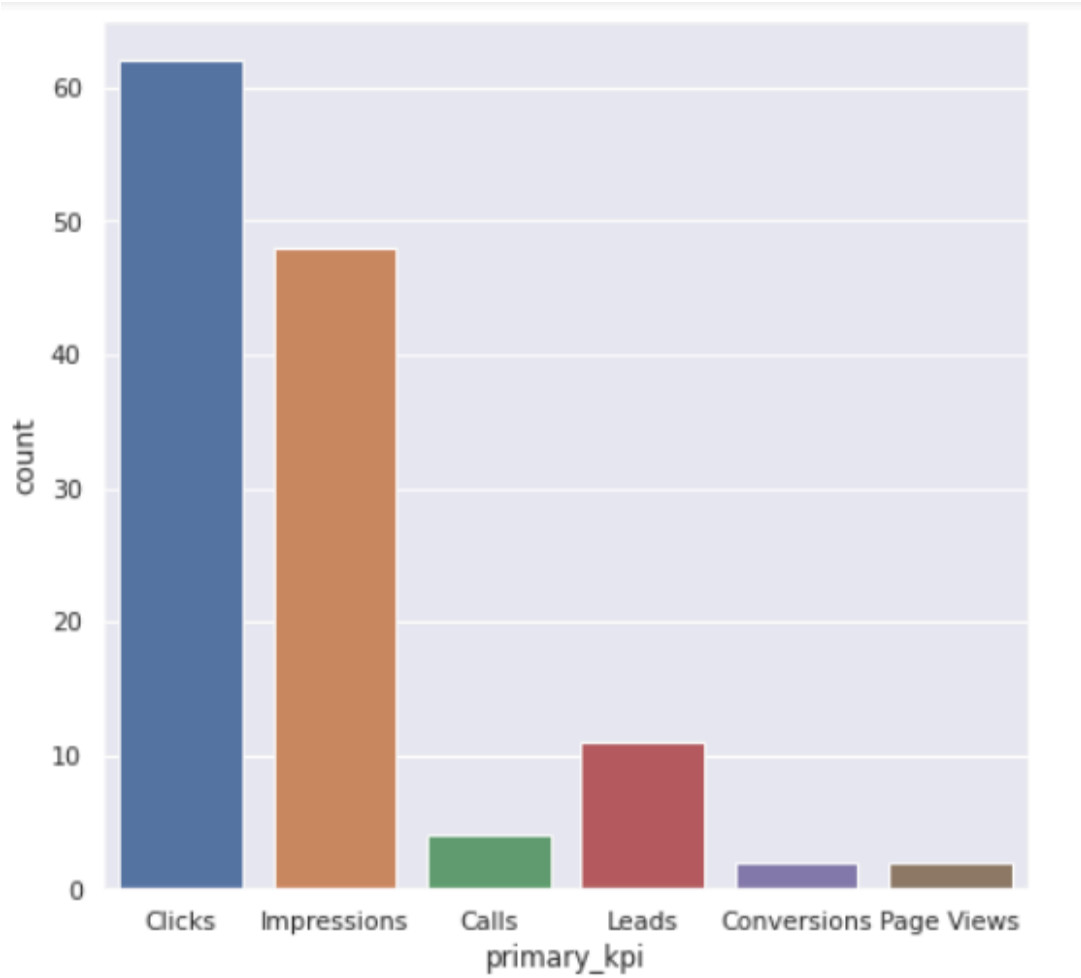

Figure 3b

From the overall result, it could be seen that the boosting algorithms out performed other machine learning algorithms and this can be attributed to the compounding effectiveness of these algorithms through their scalable tree boosting system as highlighted by *Chen and Chen (2016)*. A critical look into the f1 score, recall and accuracy showed that the classes with the highest number of records had higher performance results. This indicates and correlates to the fact that 146 number of records was not sufficient enough for the model to properly learn the patterns associated with data. This finding could also be generalised for both the classification and regression approaches used in this work.

Recommendation and Conclusion

In conclusion, more attention should be given to phone call making to generate and increase student recruitment/leads for the University of Hull. This act of phone call making (also called cold-calling) is a subset of telemarketing; a process of using the telephone to generate or increase leads, sales, or gather marketing information. Phone-calling is a traditional tool used for generating leads but has been overtaken by the new emergence of social media and some

criticism/skepticism have surrounded the effectiveness of cold-calling today. However, with COVID-19, research shows that companies that apply a multichannel strategy in which they combine offline and online channels, perform better in terms of share of wallet (Melis et al. 2016) and in terms of revenues (Pauwels and Neslin 2015). This New situation has compelled businesses to blend some new and some old marketing practices to keep businesses going. Phone-calling has therefore regained momentum in terms of effectiveness in the marketing of Universities.

While phone-calling can be a highly effective tool for generating high-quality leads, there are some tips to keep in mind in order to maximize its effectiveness. It is important to keep in mind some tips making phone-calling an effective strategy in University marketing. Such tips include:

- It is important to be interactive with the leads that you call. While it is important to have some form of a script to ensure that every important information about the University of Hull is said in the initial contact with a prospective client.
- It is important to be a bit flexible. Strong rigidness can quickly reduce call to annoyance rather than an opportunity for winning the prospect to expand its client base.
- Researching prospective students is also important. Rather than cold calling all potential leads, it is important to narrow down the list to the most promising prospects.
- Outsourcing the phone-calling to a professional team could be a consideration for this would reduce time and cost, and increase efficiency. This will also give room to rapidly generate the best leads organically.
- Taking immediate action on phone-call request; If there is a request for phone call or relevant information from a prospect, put the call through or share the information as soon as possible and do not waste time because it could lead to frustration and change of mind for the prospect.

The present marketing campaign is good but an increase in effort in phone call marketing will make it more effective.

References

Chen, T. and Guestrin, C., 2016. XGBoost: a scalable tree boosting system ACM SIGKDD international conference on knowledge discovery and data mining. *ACM*, pp.785-794.

Clark, M., Fine, M., & Scheuer, C. (2017) ‘Relationship quality in higher education marketing: the role of social media engagement’ *Journal of Marketing for Higher Education*, 27(1), 40-58

Drummond, G. (2004), 'Consumer confusion: reduction strategies in higher education', *International Journal of Educational Management*, vol., no.5, pp. -317-23

Filip, A. (2012). ‘Marketing theory applicability in higher education; *Procedia-Social and Behavioral Sciences*, 46(6), 912-916

Fuchs, Christian. (2010) ‘studiVZ: Social Networking Sites in the Surveillance Society’ *Ethics and Information Technology* , 12 (2), 171-185.

Hayes, T. (2007) ‘Delphi study of the future of marketing of higher education’ *Journal of business research*, 60(9), 927-931.

Hisel and Pinion(2020) ‘Student Recruitment and Engagement in COVID 19 era’ *Journal of Environmental Health*

Hossain, S., & Sakib, M. D. (2016) ‘The Impact of Social Media Marketing on University Students' Brand Loyalty’ *International Journal of Marketing and Business Communication,* 5(3), 1-7.

Merrill, N. (2011) *Social media for social research: applications for higher education communication*. Higher Education Administration with Social Media (Cutting- Edge Technologies in Higher Education), Emerald Group Publishing Limited, 2,25-48.

Nazeer, J. (2017) ‘Impact of Social Media Advertisements on University Students’ *Arts and Social Sciences Journal*, 8(4), 1-6

Ozbal & Konakli (2015) ‘A Literature Review to Marketing of Higher Education’ *Regional Economic Development Through Entrepreneurship and Innovation, 18-22*

Paladan, N. (2018) ‘Higher Education Institutions Embracing Digital & Social Media Marketing: A Case of Top 25 Universities in Asia & Africa’ *AIMI Journals of Marketing and Branding Research,* 5, 159-167.

Pasternak, R. (2005). Choice of institutions of higher education and academic expectations: the impact of cost-benefit factors. *Teaching in Higher Education*, *10*,(2): 189–201.

Rutter, R., Roper, S., & Lettice, F. (2016) ‘Social media interaction, the university brand and recruitment performance’ *Journal of Business Research*, 69(8), 3096-3104.

Świeczak & Lukowski (2016), ‘Lead generation strategy as a multichannel mechanism of growth of a modern enterprise’, *Marketing of Scientific and Research Organizations,*

Appendix 1.